

Secondhand AI

Ambient AI scribes now document millions of patient visits a year [1]. They listen to the consultation and draft a clinical note. A treating clinician may read the note, sign it, and enter it into the medical record. After that, AI’s role is invisible. A different clinician who later opens the patient’s chart may read the note as the judgment of her colleague. She has no way to tell whether the treating clinician wrote it carefully or approved an AI draft with minimal changes. A patient who reads a follow-up note through her portal sees her doctor’s name; any involvement of an AI scribe is typically unclear. In one instance, a scribe invented a medication that the patient had not been prescribed [2]. Neither the downstream clinician nor the patient used AI, yet both are affected by it. Both judge the note by how it reads in a world where AI has changed what fluency means.

We call this **secondhand AI**: the downstream effect of one person’s AI use on what others receive, how they interpret it, and what they pass on. A student learns from a classmate who consulted ChatGPT. A manager receives more AI-generated work from her analysts than she can meaningfully review. In each case, the person affected by AI may not have used it, and may not even know that it was involved.

Secondhand AI, however, can also benefit the recipient. A parent who uses AI to understand a child’s diagnosis may ask better questions at the next appointment. A student who uses AI to find gaps in their reasoning may become a better tutor. The risks of secondhand AI are less visible and understudied, but they urgently need attention. While much research focuses on Alice (the person who uses AI), we ask what happens to Bob (the person who receives what she produces). We begin by briefly discussing how AI changes Alice. We then identify how Alice’s AI use reaches Bob, and propose how to measure and design for those on the receiving end.

AI changes the sender

AI can help Alice perform better in the moment yet risk keeping her from learning deeply. For example, high school mathematics students given access to standard GPT-4 performed 48% better during AI-assisted practice but scored 17% worse on subsequent unassisted exams [3]. Students used the AI assistant as a “crutch,” leaning on it instead of doing the work required to acquire the skill. As performance was high when AI was used, students had little reason to suspect their understanding was shallow. Messeri and Crockett [4] call this an *illusion of understanding*: Alice believes she understands more than she does because AI filled gaps she would otherwise have had to confront. The illusion is a first-order problem for Alice. It becomes a secondhand AI problem the moment she explains what she learned to Bob.

What reaches Bob depends partly on how carefully Alice checks her own work. A survey found that knowledge workers who expressed higher confidence in generative AI were less likely to critically evaluate its outputs before passing them on [5]. At the same time, AI raises the floor of quality. A generative AI assistant increased productivity of customer-support workers by 15% on average, with a 34% improvement for the least experienced workers [6]. As AI narrows the gap between novices and experts, Bob can no longer use fluency to gauge how much Alice knows.

Three modes of secondhand AI

Alice’s AI use can reach Bob through the content she produces or through who she becomes. The former creates problems of volume and signal quality, while the latter is harder to see. We describe three modes below, following the ambient AI scribe through each.

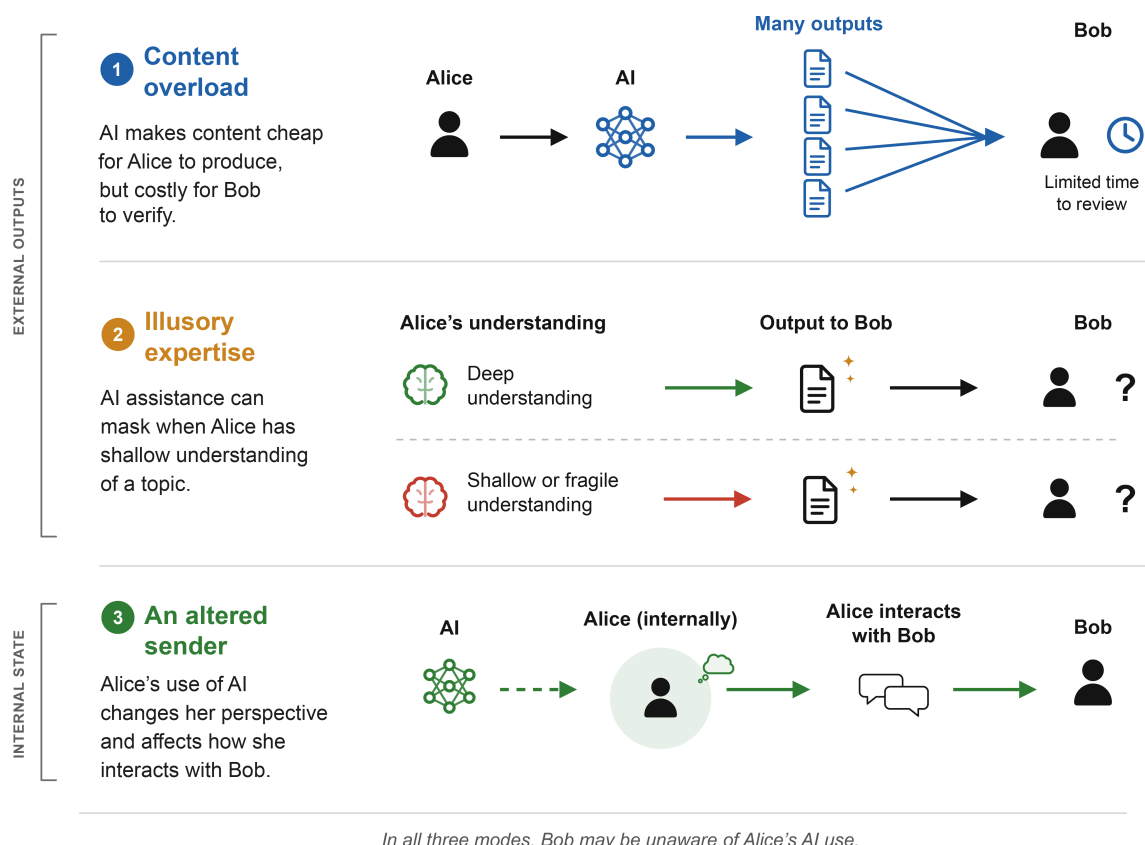


Figure 1: Three modes by which Alice’s use of AI can reach Bob: (1) content overload when Bob receives material from Alice, (2) illusory expertise present in Alice’s outputs, and (3) an altered sender (Alice) who is shaped by AI prior to social interactions.

Content overload

Today, AI makes content cheap to produce but costly to verify. The time Alice saves generating it is time Bob loses reviewing it. Before AI, effort limited output. That limit roughly matched the capacity of people downstream. AI lowers the cost of content production. The volume of content that reaches Bob grows while his capacity to review it stays the same unless he uses AI.

AI-generated clinical notes, as of 2026, tend to be longer than physician-authored ones [1]. When an AI scribe produces a note for every visit, the next clinician faces more to read with no more time to read it. To the extent that a physician-authored note served as a record and evidence of cognitive engagement with the case, an AI-drafted note weakens that assurance. The next clinician must choose between verifying the note and seeing the next patient, yet may not have time for both.

The same dynamic appears in knowledge work. In a survey of 1,150 US desk workers conducted by researchers at Stanford and BetterUp Labs, 40% reported receiving AI-generated low-quality work product, or “workslop,” in the previous month [7]. Each instance cost the recipient nearly two hours to interpret or repair. The authors estimated this verification burden

at \$186 per employee per month in lost productivity or more than \$9 million per year for a 10,000 person organization. 42% of recipients subsequently viewed the sender as less trustworthy. When Bob cannot verify what he receives from Alice, he either trusts it uncritically or discounts everything: neither serves him well. To keep up with Alice’s output, Bob may be pressured to adopt AI in his review process. Her AI use compels his.

Illusory expertise

AI acts as a mediator between Alice’s knowledge and her output, which can weaken the link between appearing knowledgeable and being knowledgeable. Before AI, linguistic fluency in a document carried information. A polished note suggested the clinician had spent time on it. Those linguistic signals were costly to produce. AI lowers the cost of fluency and can produce confident prose irrespective of the underlying reasoning. Since Bob judges Alice by what she produces, he has now lost a reliable cue.

In the case of an ambient scribe, a note in the medical record may appear to reflect authoritative judgment irrespective of how much time the clinician spent on it. AI may replace the clinician’s shorthand with fluent prose that simulates professional judgment. Patients rate AI-generated clinical messages more favorably when they seem polished, but reverse that preference once told AI was involved [8]. This pattern extends beyond medicine. An AI-assisted legal filing may read as competent legal work but may contain fabricated citations. The cost of discovering those errors falls to opposing counsel and judges who never use AI themselves.

Fluency obscures weak reasoning and makes language more persuasive. For example, Salvi et al. [9] found that GPT-4 changed people’s views just as much as human debaters. The AI assistant became more persuasive when it was given sociodemographic information about the listener. The polish of Alice’s language may mask Bob’s ability to tell how much she understands.

An altered sender

Bob may interact with Alice without realizing that AI shaped her judgment. Jakesch et al. [10] found that people who used an opinionated AI writing assistant were more likely to report holding the AI’s view on a subsequent survey. Their opinion shifted during their AI interaction without them noticing.

A clinician who uses an ambient AI scribe may find that it colors how she frames cases. When she briefs a colleague, her framing and language may reflect her AI assistant. Beyond healthcare, a student who learns a concept with an AI assistant or a parent who researched her child’s diagnosis with a chatbot may carry the influence of AI into their subsequent conversations.

This mode is hardest to address. Disclosure would have helped if there was an artifact to label. Here, however, AI’s contribution is how Alice *thinks* and what she recalls and believes. Neither she nor Bob may know which of her ideas are her own and which were shaped by the AI she used earlier.

Measuring and designing for the recipient

Secondhand AI becomes visible only when researchers study what happens *between* people. Current evaluation frameworks tend to measure first-order outcomes: task completion, satisfaction, persistence or memory. They do not measure what happens to Bob and all the others with whom Alice interacts. Three measurement mechanisms can fill this gap. *Dyadic experiments* randomly assign Alice to use AI or not, then measure what Bob learns from her output. Outcomes should include Bob’s accuracy, confidence, trust, verification effort, and ability to transmit the knowledge onward. *Transmission-chain experiments* track how content degrades as it passes from Alice to Bob, and then to Carol and Dave. *Institutional audits* can examine workflows: whether

content can be traced back to its source, how much downstream review it requires, and where AI involvement is lost.

For content overload, the priority is to measure how Bob’s verification behavior shifts as the amount of AI-generated content grows. The workslop survey used self-reports to establish that the burden of review is real [7]. Controlled experiments could vary the proportion of AI-generated content in Bob’s workflow to measure whether his scrutiny drops as volume rises and at what threshold he shifts from checking carefully to blindly trusting AI outputs.

For illusory expertise, dyadic experiments can isolate the secondhand AI effect directly. Cavalier et al. [8] suggest that disclosing AI involvement is meaningful, but its effect on calibration is unknown. One way to test this: randomly assign medical students to prepare a case summary with or without AI assistance. Give each summary to a second student who is not told whether AI was involved. Measure whether AI-assisted summaries produce higher confidence in the second student without improving her actual understanding. Then, vary disclosure: does telling the second student that AI may have been involved restore her calibration?

For an altered Alice, measurement must also focus on human-human interaction. Alice can learn a concept with or without AI and then teach it to Bob. Measure whether Bob can get the answer correct, explain his reasoning, and handle edge cases. Does Bob recognize the limits of what he has been told? Transmission-chain experiments test the limits of understanding as it passes from link to link: is Bob able to explain a concept to Carol well enough that Dave will understand too?

Design interventions differ depending on whether Bob is receiving Alice’s output or talking to a changed Alice. In the scribe case, this could mean flagging sections of a signed clinical note that the clinician edited heavily versus those approved unchanged, so the next clinician can calibrate her reading accordingly. Legal marking rules, such as the EU AI Act’s transparency provisions, are a start because they ask whether AI was involved. Secondhand AI requires more: enough information for the next person to scrutinize what they receive. If AI use changes Alice, institutions may move beyond disclosure of use and check whether Alice still has the ability to exercise independent judgment. In practice, some institutions may ask Alice to demonstrate her judgment unaided, so that any erosion in her abilities do not affect Bob.

Secondhand AI names the space where one person’s AI assistance becomes another person’s epistemic environment. Those recipients did not choose to use AI, may not know AI was involved, and often lack a way to contest or correct the effect. Secondhand smoke is an imperfect analogy, as the potential benefits of AI exposure should be considered alongside the potential harms. But the structural parallel is helpful: the costs of AI use do not stay with the user. They travel through the social channels that carry knowledge, trust, and professional judgment. Whether we detect secondhand AI and its effects in time is a question our field has barely begun to ask.

References

- [1] Paul J Lukac, William Turner, Sitaram Vangala, Aaron T Chin, Joshua Khalili, Ya-Chen Tina Shih, Catherine Sarkisian, Eric M Cheng, and John N Mafi. Ambient AI scribes in clinical practice: a randomized trial. *NEJM AI*, 2(12):AIoa2501000, 2025. 1, 2
- [2] Office of the Auditor General of Ontario. Use of Artificial Intelligence in the Ontario Government. Special Report, Office of the Auditor General of Ontario, Toronto, ON, Canada, May 2026. URL https://www.auditor.on.ca/en/content/specialreports/specialreports/en26/2026_AI_EN.pdf. 1
- [3] Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman. Generative ai without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26):e2422633122, 2025. 1
- [4] Lisa Messeri and Molly J Crockett. Artificial intelligence and illusions of understanding in scientific research. *Nature*, 627(8002):49–58, 2024. 1

- [5] Hao-Ping Lee, Advait Sarkar, Lev Tankelevitch, Ian Drosos, Sean Rintel, Richard Banks, and Nicholas Wilson. The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers. In *Proceedings of the 2025 CHI conference on human factors in computing systems*, pages 1–22, 2025. [1](#)
- [6] Erik Brynjolfsson, Danielle Li, and Lindsey Raymond. Generative AI at work. *The Quarterly Journal of Economics*, 140(2):889–942, 2025. [1](#)
- [7] Kate Niederhoffer, Gabriella Rosen Kellerman, Angela Lee, Alex Liebscher, Kristina Rapuano, and Jeffrey T Hancock. AI-generated “workslop” is destroying productivity. *Harvard Business Review*, 2025. [2](#), [4](#)
- [8] Joanna S Cavalier, Benjamin A Goldstein, Vardit Ravitsky, Jean-Christophe Bélisle-Pipon, Armando Bedoya, Jennifer Maddocks, Sam Klotman, Matthew Roman, Jessica Sperling, Chun Xu, et al. Ethics in patient preferences for artificial intelligence–drafted responses to electronic messages. *JAMA Network Open*, 8(3):e250449, 2025. [3](#), [4](#)
- [9] Francesco Salvi, Manoel Horta Ribeiro, Riccardo Gallotti, and Robert West. On the conversational persuasiveness of GPT-4. *Nature Human Behaviour*, 9(8):1645–1653, 2025. [3](#)
- [10] Maurice Jakesch, Advait Bhat, Daniel Buschek, Lior Zalmanson, and Mor Naaman. Co-writing with opinionated language models affects users’ views. In *Proceedings of the 2023 CHI conference on human factors in computing systems*, pages 1–15, 2023. [3](#)